Patterns of Truth

Location and Modality

Gabriel Uzquiano · University of Southern California

1. Patterns of Truth

Tense logic takes propositions to change truth value across times. Consider the proposition expressed by (1):

(1) There is a major US election.

This proposition is false now, but it will be true on November 3, 2026. Temporary propositions may be true:

- all over a time interval
- entirely within a time interval
- at some of a time interval

Or you may take propositions to change truth value across possible worlds. Consider the proposition expressed by (2):

(2) Someone walked on the Moon in 1969.

This proposition is true in the actual world, but it might have been false had the political will been absent. Contingent propositions may be true:

- all over a region of modal space
- entirely within a region of modal space
- at some of a region of modal space

We aim for a generalization. We leave open whether locations are time intervals, or regions of modal space, or regions of spacetime, what matters is that they enter into the relation of part to whole.

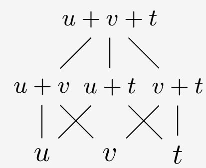
2. Locational Frames

A **locational frame** consists of

- a non-empty set L of locations, and
- a relation of **superlocation** $\sqsupseteq$,

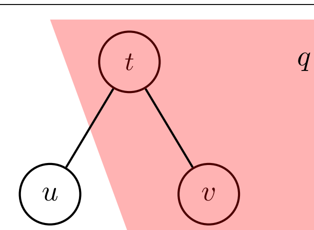
where $u \sqsupseteq v$ means that v is part of u.

$\sqsupseteq$ is reflexive, anti-symmetric, and transitive.



Example. Let u, v, and t be Spring, Summer, and Fall. Spring and Summer lie below Spring + Summer because they are subintervals of it.

Truth at a location. We begin with the relation an atomic proposition bears to a location if it is true *all over* that location.



$V(q) = \{t, v\}$

Heredity. If an atom is true at a location, then it remains true at every sublocation.

3. Locational Modalities

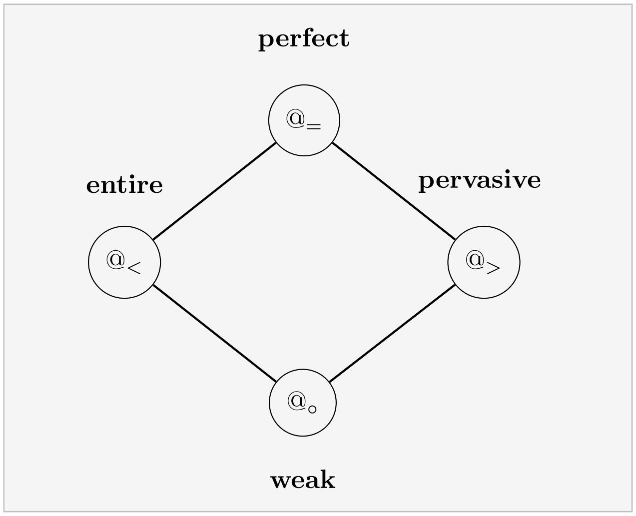
Two basic modal operators:

$u \Vdash \Box\varphi$ for all $v \in L$, if $u \sqsupseteq v$, then $v \Vdash \varphi$.
all sublocations verify φ
φ is true **all over** u

$u \Vdash \boxbar\varphi$ for all $v \in L$, if $u \sqsupseteq v$, then $u \sqsupseteq v$.
only sublocations verify φ
φ is true **only within** u.

We now characterize four locational modalities in terms of $\Box$ and $\boxbar$:

name	**formal**	**interpretation**
pervasive	$\Box\varphi$	true all over u
weak	$\Diamond\varphi$	true in some of u
entire	$\Diamond\varphi \wedge \boxbar\varphi$	true within u
perfect	$\Box\varphi \wedge \boxbar\varphi$	true all over *and* only within u



4. Exact Location

Perfect location tracks *where a proposition holds* in the most precise way the pattern of truth allows. But there is a further notion: **exact location**, tracking *what makes a proposition true.*

Consider proposition (1) again. US presidential elections occur every four years, so there are two relevant election days near the present:

- November 5, 2024 — one **truthmaker**
- November 7, 2028 — another truthmaker

Proposition (1) is *exactly located* at each election day, because each is a concrete event that makes it true. Yet (1) is **not** perfectly located at either day alone — since the other election day also makes (1) true, the proposition "spills outside" each one.

Exact location is not additive. Consider:

> *(2) The sphere completes one full rotation.*

Suppose the sphere rotates once in (t_1, t_2) and once in (t_2, t_3). Then:

- (2) is exactly located at (t_1, t_2)
- (2) is exactly located at (t_2, t_3)
- (2) is **not** exactly located at (t_1, t_3), which contains *two* rotations, not one

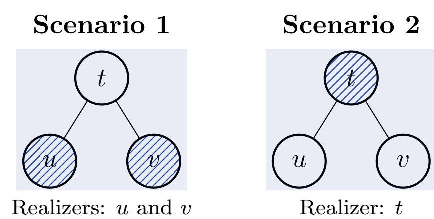
Exact location tracks individual truthmakers, not just regions where the proposition holds.

5. Exact Location is Undefinable

Main result. Exact location cannot be defined in terms of the truth patterns $\Box$ and $\boxbar$ alone.

Why? Two propositions can agree on *every* truth-pattern fact — the same locations verify φ, the same verify $\Box\varphi$, the same verify $\boxbar\varphi$ — yet differ in their exact locations.

Illustration. Let p hold at locations u, v, and t (where t contains u and v as parts):



The truth pattern is identical in both scenarios. Yet in Scenario 1, p is exactly located at u and v (two truthmakers); in Scenario 2, p is exactly located at t (one truthmaker). No formula built from $\Box$ and $\boxbar$ can tell them apart.

To capture exact location we must supplement our framework with a **realization relation**: a mapping from propositions to the locations of their truthmakers. This goes beyond what any pattern of truth can encode.

6. The Upshot

There is **more to where a proposition is true** than its pattern of distribution across locations. Exact location is partly a matter of *what makes it true* — and that cannot be read off from truth alone.

Summary Table

Notion	What it tracks	Key property
Pervasive	Truth throughout	Closed downward
Weak	Truth somewhere	Weakest
Entire	Truth confined within	—
Perfect	Both above	Unique (functional)
Exact	Truthmakers	Not truth-definable

Selected references. Casati & Varzi, *Parts and Places* (MIT Press, 1999); Correia, "Location by proxy" (ms., 2025); Li & Wang, "Mereological bimodal logics", *Review of Symbolic Logic* (2022); Parsons, "Theories of location", *Oxford Studies in Metaphysics* (2007).